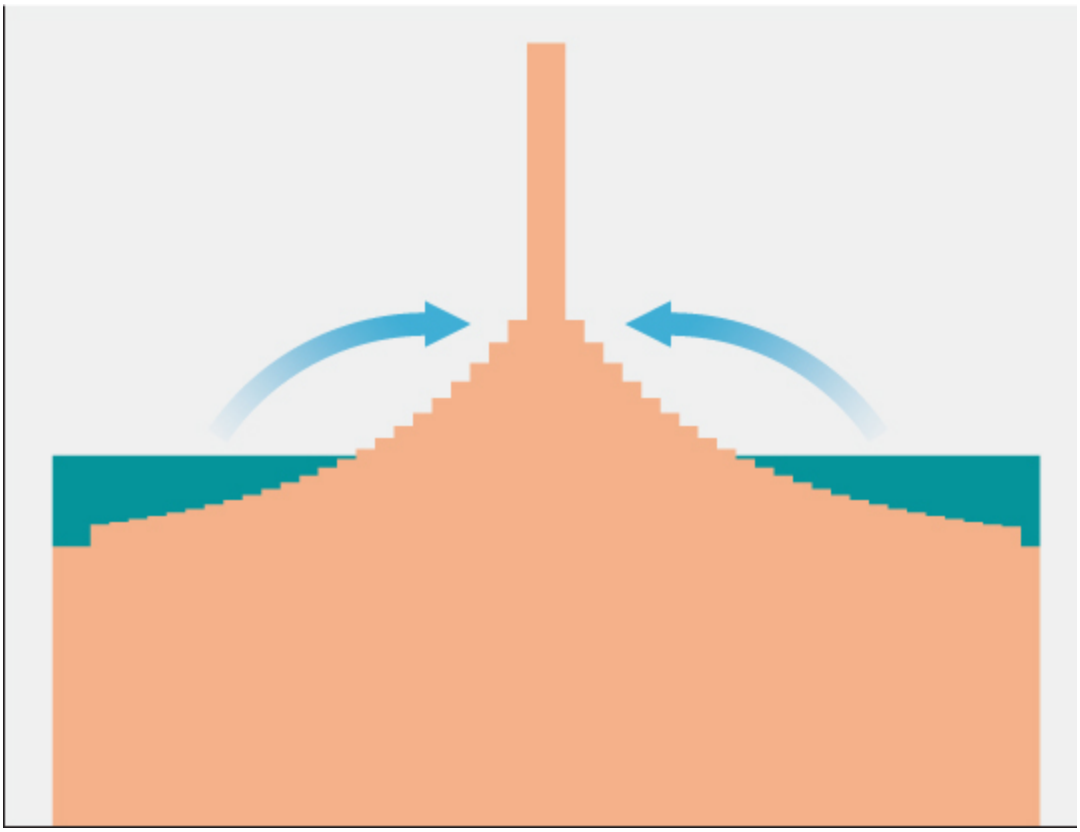


In listing 7.2 we implement the function that takes a set of supports, the associated probabilities, and an observation, and returns an updated probability distribution by redistributing the probability mass toward the observed value.

Listing 7.2. Updating a probability distribution

```
def update_dist(r,support,probs,lim=(-10.,10.),gamma=0.8):
    nsup = probs.shape[0]
    vmin,vmax = lim[0],lim[1]
    dz = (vmax-vmin)/(nsup-1.)              1
    bj = np.round((r-vmin)/dz)              2
    bj = int(np.clip(bj,0,nsup-1))          3
    m = probs.clone()
    j = 1
    for i in range(bj,1,-1):                4
        m[i] += np.power(gamma,j) * m[i-1]
        j += 1
    j = 1
    for i in range(bj,nsup-1,1):            5
        m[i] += np.power(gamma,j) * m[i+1]
        j += 1
```